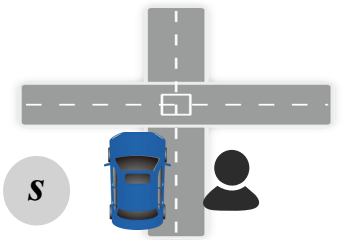


Same-State Veto Pair



Rejected
Auto Proposal

Executed
Human Action

a_n

a_h



$a_h \succ_s a_n$

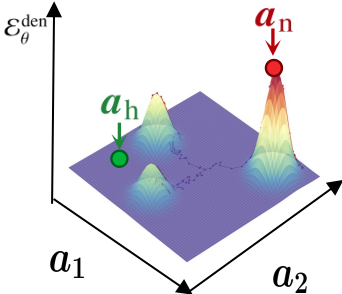
Observed Same-State Veto

Online Repair under Supervisor Authority

Diffusion Actor

Actor-Side Veto Ordering

lower = higher denoising compatibility



$$\mathcal{E}_\theta^{\text{den}}(s, a_h) < \mathcal{E}_\theta^{\text{den}}(s, a_n)$$

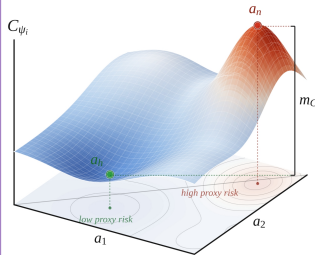
Actor-Side Learning

- Diffusion BC on Executed Actions
- EnergyRank same-state ordering
- Critic-mediated improvement

Twin Proxy-Risk Critics

Critic-Side Veto Ordering

lower = lower predicted proxy risk



$$C_{\psi_i}(s, a_h) \leq C_{\psi_i}(s, a_n)$$

Critic-Side Learning

- TD cost propagation
- CPCal per-head margin
- Pointwise PV anchoring

Frozen Guidance and Gated Execution

Diffusion Actor

θ frozen



Twin Critics

$\psi_1 \psi_2$ frozen

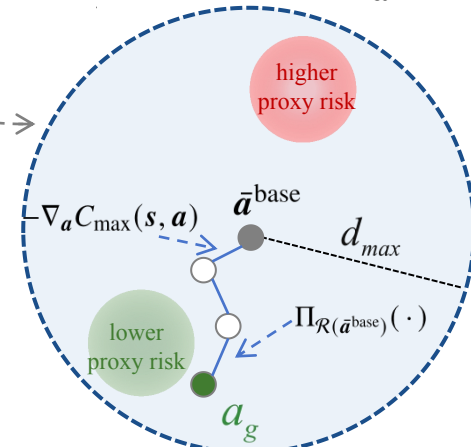
actor sample

project

a^{base}

$\bar{a}^{\text{base}}$

shared
noise



$$\mathcal{R}(\bar{a}^{\text{base}}) = \{a \in \mathcal{A}_{\text{valid}} : \|a - \bar{a}^{\text{base}}\|_2 \leq d_{\text{max}}\}$$